

Vinh Nguyen

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EDUCATION

University of Minnesota - Twin Cities, Minneapolis, MN

M.S. Computer Science, Expected Dec 2025 | B.S. Computer Science, Dec 2024 | GPA: 3.8/4.0

Relevant Coursework: Operating Systems, Embedded Systems, Secure Software, Parallel Computing, Deep Learning

WORK EXPERIENCE

Amazon Web Services (AWS) – Software Development Engineer Intern

May 2025 - August 2025

- Refactored Aurora DSQL data plane front-end components to support query processing build permutations, enabling sister teams to test experimental features in real infrastructure.
- Delivered 10 pull requests (1000+ lines of code in Rust) covering refactoring, feature development, and bug fixes, all reviewed and merged into mainline with full CI/CD validation.
- Collaborated with query processing engineers to define requirements, integrate feedback, and drive adoption of new features in testing environments.
- Finished project's core functionalities 3 weeks ahead of deadline; demoed final project to 60+ engineers org-wide, receiving direct recognition from the Director of DSQL.

Hennepin County IT Department – Software Development Intern

January 2023 - May 2024

- Built and deployed 4 full-stack apps (Angular/.NET) serving 5,000+ users across multiple departments.
- Implemented 20+ new features per sprint in a 12-member agile team, contributing to iterative releases and user-driven improvements.
- Migrated legacy apps with modern UIs and robust APIs; supported secure cloud hosting on Azure (99.9% uptime).

Small Satellite Research Lab (UMN) – Embedded Systems Developer

September 2021 - April 2022

- Programmed, tested, and debugged SPI/I²C communication protocols and embedded firmware in C++ for 5 Raspberry Pi and microcontroller platforms.
 - Coordinated with a 5-member team in weekly sprints, achieving 95% on-time task completion for project milestones.
 - Designed modular firmware enabling flexible deployment of satellite subsystem prototypes.
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PROJECTS

Rust Web Crawler

github.com/NguyenLTVinh/rust-web-crawler

- Built a parallel Rust crawler (tokio async) scaling to 5K pages/run, with deduplication and crawl queues.
- Supported static (request) + dynamic (headless Chrome) scraping for JS-heavy sites.
- Stored and indexed pages with SQLite + FTS5 for fast full-text search.

Two-Factor Authenticated Door Lock System

github.com/NguyenLTVinh/2FA-Lock-System

- Developed a secure RFID and PIN-based door lock system with Bluetooth communication.
 - Integrated AVR microcontrollers, cryptography chips, and LCD/keypad interfaces.
 - Enforced reliability with modular design and anti-forgery mechanisms.
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SKILLS

Programming languages: Rust, C, C++, C#, Python, JavaScript, TypeScript, Java

Tools: Linux (Debian based), Cargo, CMake, Docker, SQL, Git, MATLAB, Anaconda

Frameworks: React.js, Angular, .NET, Express, TensorFlow, PyTorch, OpenCV